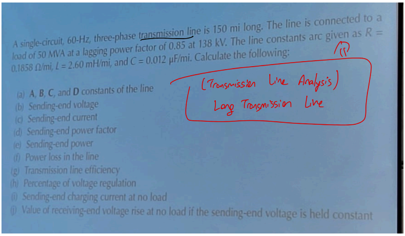


A single-circuit, 60-Hz, three-phase transmission line is 150 mi long. The line is connected to a load of 50 MVA at a lagging power factor of 0.85 at 138 kV. The line constants arc given as $R = 0.1858\ \Omega/\text{mi}$, $L = 2.60$ mH/mi, and $C = 0.012\ \mu\text{F/mi}$. Calculate the following:

(a) **A**, **B**, **C**, and **D** constants of the line
(b) Sending-end voltage
(c) Sending-end current
(d) Sending-end power factor
(e) Sending-end power
(f) Power loss in the line
(g) Transmission line efficiency
(h) Percentage of voltage regulation
(i) Sending-end charging current at no load
(j) Value of receiving-end voltage rise at no load if the sending-end voltage is held constant

(Transmission Line Analysis)
Long Transmission Line

$$V_S = AV_R + BI_R$$

$$I_S = CV_R + DI_R$$

$$Z = R + jwL \qquad \gamma = \sqrt{zy}$$

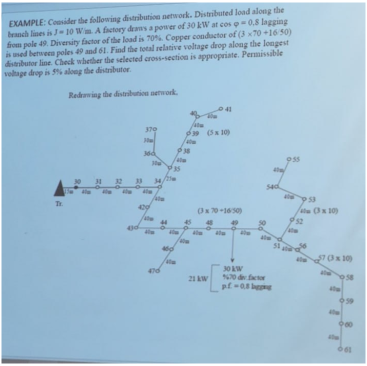
$$y = jwC \qquad Z_c = \sqrt{\frac{z}{y}}$$

$$A = D = \cosh(\gamma l)$$

$$B = Z_c \sinh(\gamma l)$$

$$C = \frac{1}{Z_c} \sinh(\gamma l)$$

HW sorusu
en sona
çok uzun
bak



EXAMPLE: Consider the following distribution network. Distributed load along the branch lines is $J = 10$ W/m. A factory draws a power of 30 kW at $\cos\varphi = 0.8$ lagging from pole 49. Diversity factor of the load is 70%. Copper conductor of $(3 \times 70 + 16/50)$ is used between poles 49 and 61. Find the total relative voltage drop along the longest distributor line. Check whether the selected cross-section is appropriate. Permissible voltage drop is 5% along the distributor.

Redrawing the distribution network.

1) $\%\Delta U_{max} = \%5$ → izin verilen max sınır.

Yayılı yük: $j = 10$ W/m

49 nolu direkte fabrika: $P = 30$ kW
$\cos\phi = 0{,}8$ lagging

Diversity factor: %70 = 0,70

Bu yüzden fabrikanın aktif gücü : $P_f = 30\,000 \times 0,70$

$P_f = 21\,000$ W

2) En uzun yolu seçiyoruz.

Tr → 30 → 34 → 45 → 49 → 51 → 61

TR-30	15m
30-34	160 m
34-45	160 m
45-49	80 m
49-51	80 m
51-61	240 m

3) Yayılı yükleri noktasal yüke çeviriyoruz.

10 W/m

30 nok. yük

30-34 arası 160 m

$160 \times 10 = 1600$ W

Yarısı 30. noktaya
" 34. " gelir

$P_{30} = 800$ W

34 nok. yük

30-34 hattının yarısı ; $P_{34} = 800$ W

34-45 " " ; 800W

34'e bağlı yan yol ; $245\text{m} \times 10 = 2450$ W

Toplam;

$P_{34} = 800 + 800 + 2450$

$P_{34} = 4050$ W

45 noktasındaki yük;

34-45 hattının yarısı; 800 W

45-49 " " ; 400 W

45'e bağlı yan kol; $80\text{m} \times 10 = 800$ W

$P_{45} = 800 + 400 + 800$

$= 2000$W

49 noktasındaki yük;

45-49 hattının yarısı; 400 W

49-51 " " ; 400 W

fabrika yükü : 30kW × 0,7 = 21 000 W

$P_{49} = 21\,800\,W$

51 noktasındaki yük;

49-51 hattının yarısı; 400 W

51-61 hattının yarısı; 1200 W

51'e bağlı yan kol; 160m × 10 = 1600 W

$P_{51} = 3200\,W$

61 nok. yük;

51-61 hattı 240 m;

$240 \times 10 = 2400\,W$

Yarısı; P_{61} : 1200 W

Her hat parçasının taşıdığı gücü buluyoruz.

→ 51-61 arası;

$P = 1200\,W$

→ 49-51 arası;

51 ve 61 yüklerini taşır.

$P = 3200 + 1200 = 4400W$

→ 45-49 arası;

49, 51, 61 yüklerini taşır.

$P = 21800 + 3200 + 1200$

→ 34-45 arası;

45, 49, 51, 61 yüklerini taşır.

$P = 2000 + 21800 + 3200 + 1200$

$= 28200W$

→ 30-34 arası;

34, 45, 49, 51, 61 yükl. taşır.

$P = 4050 + 28200\ W$

$P = 32250\,W$

→ Tr-30 arası;

Tüm yükleri taşır;

$P = 800 + 32250$

$= 33050$

Son hali

Yük çizimi

$$\text{Trafo} \overset{15m}{\text{———}} 30 \overset{160m}{\text{———}} 34 \overset{160m}{\text{———}} 45 \overset{80m}{\text{———}} 49 \overset{80m}{\text{———}} 51 \overset{240m}{\text{———}} 61$$

Hatların taşıdığı güç çizimi

$$\text{Trafo} \underset{33050W}{\overset{15m}{\text{———}}} 30 \underset{32250W}{\overset{160m}{\text{———}}} 34 \underset{28200W}{\overset{160m}{\text{———}}} 45 \underset{26200W}{\overset{80m}{\text{———}}} 49 \underset{4400W}{\overset{80m}{\text{———}}} 51 \underset{1200W}{\overset{240m}{\text{———}}} 61$$

Permissible Relative Volt. Drops.

→ Indoor lightning network : %ε = 1,5
→ low volt. power network (including motors) : %ε = 3
→ Distribution network : %ε = %3 - %5 (önceki sorudan %5 kullandık)
→ Medium volt. lines : %ε = %5 - %10

Türkiyede

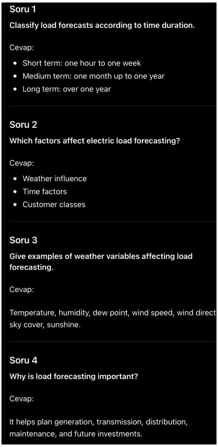
Low Volt. Levels : 220, 380
Medium Volt. " : 34,5 kV
High Volt. " : 154 kV
Extra High Volt. " : 400 kV

What factor affect the selection of conductor cross-section?

→ Current carrying capacity
→ Power loss
→ Voltage drop
→ Mechanical strength

If the distance between source and load is short, which factor is most important?

→ Current capacity of conductor



Soru 1

Classify load forecasts according to time duration.

Cevap:

- Short term: one hour to one week
- Medium term: one month up to one year
- Long term: over one year

Soru 2

Which factors affect electric load forecasting?

Cevap:

- Weather influence
- Time factors
- Customer classes

Soru 3

Give examples of weather variables affecting load forecasting.

Cevap:

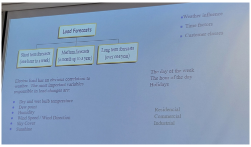
Temperature, humidity, dew point, wind speed, wind direct
sky cover, sunshine.

Soru 4

Why is load forecasting important?

Cevap:

It helps plan generation, transmission, distribution, maintenance, and future investments.



Load Forecasts

Short term forecasts (one hour to a week)

Medium forecasts (a month up to a year)

Long term forecasts (over one year)

- Weather influence
- Time factors
- Customer classes

Electric load has an obvious correlation to weather. The most important variables responsible in load changes are:

- Dry and wet bulb temperature
- Dew point
- Humidity
- Wind Speed / Wind Direction
- Sky Cover
- Sunshine

The day of the week
The hour of the day
Holidays

Residencial
Commercial
Industrial

Disadvantages of low power factor

$$I_L = \frac{P_L}{\sqrt{3} V \cos\phi}$$

$\cos\phi$ = power factor
P_L = aktif güç
V = hat gerilimi

$pf\downarrow$

1) Generator and transformer rating ↑

Aynı aktif gücü vermek için daha büyük kVA / MVA kapasiteli jeneratör ve transformatör gerekir.

$$\boxed{S = \frac{P\uparrow}{pf\downarrow}}$$

S = görünür güç
P = aktif güç
Q = reaktif güç

2) Hatlar ve kablolar daha fazla akım taşır.

When pf decreases, current is increases. That's why cable cross section area and cost is increase.

3) Güç kaybı artar.

$$\underset{\uparrow}{P_{loss}} = \underset{\uparrow}{I^2} \cdot R$$

Akım arttığı için güç kaybı da artar. Ve efficiency is decreases

4) voltage drop is decrease

* How do you fix low power factor?

shunt capacitor
(paralel capacitörleri bağlayarak)

capacitor $\Rightarrow$ θ verir $\rightarrow$ leading $\rightarrow$ önde $(-\theta)$

inductance load $\Rightarrow$ θ $\overset{\text{reactive güç}}{\text{tüketir.}}$ $\Rightarrow$ lagging (gecikme) $(+\theta)$

* low pf reason $\Rightarrow$ inductive load (pf ↓)

low pf solution $\Rightarrow$ capacitor (pf ↑)

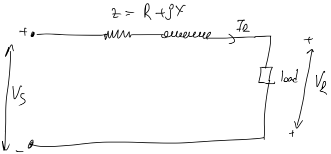
Short line model practice Question

A short transmission line has the following parameters:

- Series impedance: Z = 4 + j8 Ω
- Receiving-end voltage: V_R = 11,000 V (take as reference: V_R = 11000∠0°)
- Load current: I_R = 100 A at power factor = 0.8 lagging
- Since it's a short line: I_S = I_R

Find:

(a) The sending-end voltage V_S
(b) The voltage regulation (%)
(c) Sketch the phasor diagram



a) $\boxed{\text{short line}}$

$V_S = V_R + z \cdot I_R$ $\qquad$ $I_S = I_R$

$V_S = 11{,}000\ V + (4 + j8) \cdot I_R$ $\qquad$ 0,8 lagging (−)

$= 11{,}000 + (4 + 8j)(100\angle -36{,}87^\circ)$

$V_S = 11\,800 + j\,400$

$\cos(\theta) = 0{,}8$

$\cos^{-1}(0{,}8) = 36{,}87^\circ$ (lagging −)

$I_R = 100\angle -36{,}87^\circ$

$I_R = 80 - j60$

b) Voltage regulation

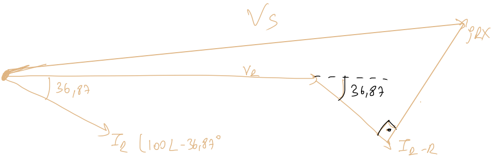
$$VR\% = \frac{|V_S| - |V_R|}{|V_R|} \times 100$$

$$V_S = \sqrt{11800^2 + 400^2} = 11{,}807{,}6$$

$$= \frac{(11800 + j400) - |11000|}{11000} \times 100$$

$$V_R = \%7{,}34$$

c)



Example) A 220kV, three phase transmission line is 60 km long. The resistance is 0,15 Ω / Km and inductance 1,4 mH / Km. Use short line model to find the voltage and power at the sending end and the voltage regulation and efficiency when the line is supplying a three phase load of assuming $f = 50\,Hz$

a) 300 MVA at 0,8 pf lagging at 220 kV

b) 300 MVA at 0,8 pf leading at 220 kV

In short line

$Z = R + jX$, $V_S = V_R + Z \cdot I_R$

$I_S = I_R = I$

$R = 0{,}15\,\Omega / km$ → km basına direnci

$R_{Top} = 0{,}15 \times 60 = \underline{9\,\Omega}$

$L = 1{,}4\,mH / Km$ → km basına mH

$L_{Top} = 1{,}4 \times 10^{-3} \times 60 = \underline{0{,}084\,H}$

$X = wL = 2\pi f L = 2\pi \cdot 50 \cdot 0{,}084 = 26{,}39\,\Omega$

$$\boxed{Z = 9 + j26{,}39\,\Omega}$$

Three phase dediği için faz başına gerilimi bulmamız lazım:

$$\boxed{V_\ell = \frac{220}{\sqrt{3}}} = 127\angle 0^\circ \text{ kV} = 127000\angle 0^\circ$$

$\cos(\theta) = 0,8$ (lagging) $\rightarrow \cos^{-1}(0,8) = \theta$

$\theta = \underline{36,87}$

for lagging $\quad I\angle -36,87^\circ$

for leading $\quad I\angle 36,87^\circ$

Akımı da bulalım:

three phase dediği için √3 'e böldük

$$I = \frac{S}{\sqrt{3}V_L}$$

$$I = \frac{300\times10^6}{\sqrt{3}\,(220\times10^3)} = 787.3 \text{ A}$$

Akım açısını yazarken logging −

for logging

$$I = 787,3\angle -36,87^\circ \text{ A}$$

$$V_S = V_\ell + I\cdot Z$$

$$= 127000\angle 0^\circ + \underbrace{\left(787,3\angle -36,87 \cdot 9+j26,39\right)}_{18261+12370j}$$

$$V_S = 145261 + j12370$$

$$V_S = \sqrt{145261^2 + 12370^2} = 145,677 \text{ V (faz)}$$

$$V_S\cdot\sqrt{3} = 145,677\cdot\sqrt{3} = 252400 \text{ V}$$

Volt. Regulation

$$VR\% = \frac{|V_S| - |V_\ell|}{|V_\ell|}\times 100 = \frac{145,677 - 127,017}{127,017}\times 100 = 14,7\%$$

$$P_{kayıp} = 3\times |I_\ell|^2 \times R = 3\times 787,3^2 \times 9 = 16,74 \text{ MW}$$

$$\eta = \frac{P_\ell}{P_S}\times 100 = \frac{240}{240+16,74}\times 100 = \underline{93,5\%}$$

b) leading

$$I = 787,3\angle 36,87^\circ$$

$$V_S = V_R + \underbrace{I_R \cdot Z}_{\downarrow} =$$

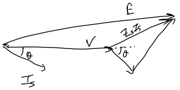
$$= V_R + \underbrace{(9+j26{,}39)(629{,}8+j472{,}4)}$$

$$= \qquad -6798 + 20872$$

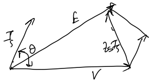
$$V_s = 120201 + j20872$$

$$V_s = \sqrt{120201^2 + 20872^2} \rightarrow 122001\,V = 122\,kV$$

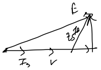
$$VR\,\% = \frac{122001 - 127000}{127000} \times 100 = -3{,}93\%$$



lagging pf load



leading pf load



unity pf load

Santral load factor & capacity factor

∝ load factor

$$\text{Load factor} = \frac{\text{Average load}}{\text{Peak load}}$$

Yükün ne kadar değişildiği

∝ Capacity factor

$$\text{Capacity factor} = \frac{\text{Average load}}{\text{Capacity}}$$

Santralin ne kadar kullandığı

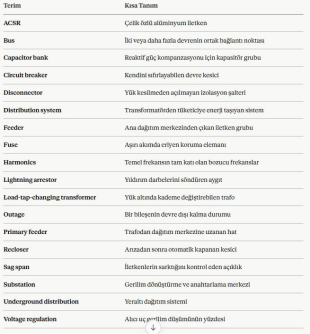
Q1) What are advantages 3 ø (phase) power over single phase

- → Can transmit more power for same amount of wire
- → Use less material for same power rating
- → Start more easily than single phase machines

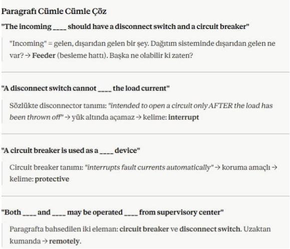
Q2) Why is reactive power supplied locally rather than transmitted over the line?

→ Because supplying reactive power locally reduces line current. Lower current means less I^2R losses, reduced voltage drop, and smaller conductor requirements.

Çıkmış boşluk doldurma → cevapları



Terim	Kısa Tanım
ACSR	Çelik özlü alüminyum iletken
Bus	İki veya daha fazla devrenin ortak bağlantı noktası
Capacitor bank	Reaktif güç kompanzasyonu için kapasitör grubu
Circuit breaker	Kendini sıfırlayabilen devre kesici
Disconnector	Yük kesilmeden açılmayan izolasyon şalteri
Distribution system	Transformatörden tüketiciye enerji taşıyan sistem
Feeder	Ana dağıtım merkezinden çıkan iletken grubu
Fuse	Aşırı akımda eriyen koruma elemanı
Harmonics	Temel frekansın tam katı olan bozucu frekanslar
Lightning arrestor	Yıldırım darbelerini söndüren aygıt
Load-tap-changing transformer	Yük altında kademe değiştirebilen trafo
Outage	Bir bileşenin devre dışı kalma durumu
Primary feeder	Trafodan dağıtım merkezine uzanan hat
Recloser	Arızadan sonra otomatik kapanan kesici
Sag span	İletkenlerin sarktığını kontrol eden açıklık
Substation	Gerilim dönüştürme ve anahtarlama merkezi
Underground distribution	Yeraltı dağıtım sistemi
Voltage regulation	Alıcı uç gerilim düşümünün yüzdesi



Paragrafı Cümle Cümle Çöz

"The incoming ____ should have a disconnect switch and a circuit breaker"

> "Incoming" = gelen, dışarıdan gelen bir şey. Dağıtım sisteminde dışarıdan gelen ne var? → **Feeder** (besleme hattı). Başka ne olabilir ki zaten?

"A disconnect switch cannot ____ the load current"

> Sözlükte disconnector tanımı: *"intended to open a circuit only AFTER the load has been thrown off"* → yük altında açamaz → kelime: **interrupt**

"A circuit breaker is used as a ____ device"

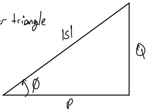
> Circuit breaker tanımı: *"interrupts fault currents automatically"* → koruma amaçlı → kelime: **protective**

"Both ____ and ____ may be operated ____ from supervisory center"

> Paragrafta bahsedilen iki eleman: **circuit breaker** ve **disconnect switch**. Uzaktan kumanda → **remotely.**

MT öncesindekiler

Power triangle



$$|S| = \sqrt{P^2+Q^2}$$

$$\phi = \tan^{-1}\left(\frac{Q}{P}\right) \qquad \frac{Q}{P} = \tan\theta$$

$$pf = \frac{P}{\sqrt{P^2+Q^2}}$$

$$S = P + jQ$$

$$|S| = \frac{P}{\cos(\phi)} = \frac{P}{pf}$$

Assume we have 100kVA load with pf = 0,8 lagging, and would like to correct the pf to 0,95 lagging.

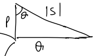
güç fak. 0,95 lagging'e yükseltilirse?

W, kW, MW	aktif güç (P)
VAr, kVAr, MVAr	reaktif güç (Q)
VA, kVA, MVA	görünür güç (S)

→ 100kVA = S

$pf = \cos\theta$

$= \frac{P}{|S|} = 0{,}8$



$\cos(\theta) = 0{,}8$

$100 \times 0{,}8 = \underline{80\,kW = P}$

$\cos^{-1}(0{,}8) = \theta$

$\theta = 36{,}87^\circ$

$\frac{\vartheta}{P} = \tan\theta$

$\frac{\vartheta}{80kW} = \tan(36{,}87^\circ)$

$\underline{\vartheta = 60\,kVar}$

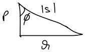
(lagging)

(kompleks gücü yazarken leading −)

$S_1 = 80 + j\,60\,kVA \qquad |S_1| = \sqrt{80^2 + 60^2} = \underline{100kVA}$

$pf_2 = 0{,}95$ lagging   P'miz aynı 80kW

$\cos^{-1}(0{,}95) = 18{,}19^\circ$



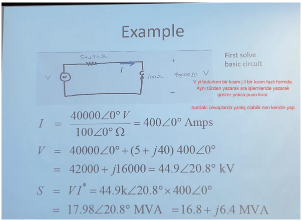
$\tan(\phi) = \frac{\vartheta}{P} = 0{,}33$

↳ 80kW

$\boxed{\vartheta = 26{,}4\,kVar}$ lag (+)

$S = P + j\vartheta$

$S = 80kW + j\,26{,}4$



Example

5+j40 Ω
I
V
100 Ω
+
40000∠0 V
−

First solve basic circuit

V yi bulurken bir kısım j li bir kısım fazlı formda. Aynı türden yazarak ara işlemleride yazarak göster yoksa puan kırar.

burdaki cevaplarda yanlış olabilir sen kendin yap

$$I = \frac{40000\angle 0^\circ\,V}{100\angle 0^\circ\,\Omega} = 400\angle 0^\circ \text{ Amps}$$

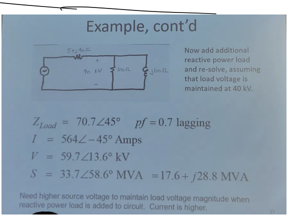
$$V = 40000\angle 0^\circ + (5 + j40)\,400\angle 0^\circ$$

$$= 42000 + j16000 = 44.9\angle 20.8^\circ \text{ kV}$$

$$S = VI^* = 44.9\text{k}\angle 20.8^\circ \times 400\angle 0^\circ$$

$$= 17.98\angle 20.8^\circ \text{ MVA} \quad = 16.8 + j6.4 \text{ MVA}$$

part 2



Example, cont'd

5+j40 Ω
+
40 kV
−
100 Ω
j100 Ω

Now add additional reactive power load and re-solve, assuming that load voltage is maintained at 40 kV.

$$Z_{Load} = 70.7\angle 45^\circ \qquad pf = 0.7 \text{ lagging}$$

$$I = 564\angle -45^\circ \text{ Amps}$$

$$V = 59.7\angle 13.6^\circ \text{ kV}$$

$$S = 33.7\angle 58.6^\circ \text{ MVA} \quad = 17.6 + j28.8 \text{ MVA}$$

Need higher source voltage to maintain load voltage magnitude when reactive power load is added to circuit. Current is higher.

Question

$V_2 = 40\angle 0^\circ\,kV$

$Z_{load} = 100\,\Omega$

$Z_{line} = 5 + j40\,\Omega$

$I = \frac{V_2}{Z_{load}} = \frac{40\angle 0^\circ}{100\angle 0} = 0{,}4\,kA\angle 0^\circ$

$V_s = V_\ell + I\,Z_{line}$

$= 40\angle 0^\circ + (2000 + j16\,000)$

$= 42\,000 + j16\,000\,V \quad = \underline{42{,}9\angle 20{,}8^\circ\,kV}$

$S_s = V_s \cdot I^*$

$I^* = 400\angle 0^\circ$ (acı 0 zaten, eşleniği aynı)

$S_s = 44{,}9\angle 20{,}8^\circ \times 400$

$= 17{,}98\angle 20{,}8^\circ$ MVA

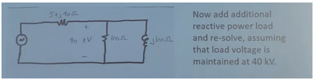
$P_{load} = V \cdot I = 40kV \times \overset{400A}{0{,}4kA} = 16MW$

$P_{loss} = I^2 \cdot R = (400)^2\,(5) = 0{,}8\,MW$

$Q_{line} = I^2 \cdot X = (400)^2\,(40) = 6{,}4\,MVar$

$\boxed{Q_{line} = 16{,}8 + j6{,}4\ MVA}$

Part 2) paralel endüktif ekleniyor



Now add additional reactive power load and re-solve, assuming that load voltage is maintained at 40 kV.

$Z_{load} = 100 \,//\, j100$ oldu

$Z_{load} = \dfrac{100(j100)}{100 + j100} = 50 + j50 = 70{,}7\angle 45^\circ$

$pf = \cos 45 = 0{,}707$

$I = \dfrac{V_\ell}{Z_{load}} = \dfrac{40\,000\angle 0^\circ}{70{,}7\angle 45^\circ} = 565{,}7\angle -45^\circ$

$\underline{I = 400 - j400\,A}$

$V_s = V_\ell + I Z_{line}$

$V_s = 40000 + (18\,000 + j14\,000)$

$\underline{V_s = 58\,000 + j14\,000\,V} \quad = 59{,}7\angle 13{,}6^\circ\,kV$

$S_s = V_s \cdot I^*$

$I^* = 400 + j400$

$S_s = (58 + j14)\,kV\,(400 + j400)\,A$

$S_s = 17{,}6 + j28{,}8$ MVA

$$\boxed{\begin{array}{l} P = 17{,}6\ MW \\ Q = 28{,}8\ MVAr \end{array}}$$